TP IaC : Yelizaveta SAMUILAVA

https://github.com/Yelizaveta-Samuilava/TP-cloud

destroy

```
yelizavetasamuilava@cloudshell:~/tp-nginx-gcp (cloud-486909)$ terraform destroy
google_compute_address.static_ip: Refreshing state... [id=projects/cloud-486909/regions/europe-west1/addresses/tp-nginx-ip]
google_compute_network.vpc: Refreshing state... [id=projects/cloud-486909/global/networks/tp-nginx-vpc]
google_compute_subnetwork.subnet: Refreshing state... [id=projects/cloud-486909/regions/europe-west1/subnetworks/tp-nginx-subnet]
google_compute_firewall.allow_ssh: Refreshing state... [id=projects/cloud-486909/global/firewalls/allow-ssh]
google_compute_firewall.allow_web: Refreshing state... [id=projects/cloud-486909/global/firewalls/allow-http]
google_compute_instance.vm_instance: Refreshing state... [id=projects/cloud-486909/zones/europe-west1-b/instances/tp-nginx-vm]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # google_compute_address.static_ip will be destroyed
  - resource "google_compute_address" "static_ip" {
      - address            = "146.148.119.178" -> null
      - address_type       = "EXTERNAL" -> null
      - creation_timestamp = "2026-03-19T03:43:58.594-07:00" -> null
      - effective_labels   = {} -> null
      - id                 = "projects/cloud-486909/regions/europe-west1/addresses/tp-nginx-ip" -> null
      - label_fingerprint  = "42WmSpB8rSM=" -> null
      - labels             = {} -> null
      - name               = "tp-nginx-ip" -> null
      - network_tier       = "PREMIUM" -> null
      - prefix_length      = 0 -> null
      - project            = "cloud-486909" -> null
      - region             = "europe-west1" -> null
      - self_link          = "https://www.googleapis.com/compute/v1/projects/cloud-486909/regions/europe-west1/addresses/tp-nginx-ip" -> nu
      - terraform_labels   = {} -> null
      - users              = [
          - "https://www.googleapis.com/compute/v1/projects/cloud-486909/zones/europe-west1-b/instances/tp-nginx-vm",
        ] -> null
    }

  # google_compute_firewall.allow_ssh will be destroyed
```

```
google_compute_firewall.allow_ssh: Destroying... [id=projects/cloud-486909/global/firewalls/allow-ssh]
google_compute_firewall.allow_web: Destroying... [id=projects/cloud-486909/global/firewalls/allow-http]
google_compute_instance.vm_instance: Destroying... [id=projects/cloud-486909/zones/europe-west1-b/instances/tp-nginx-vm]
google_compute_instance.vm_instance: Still destroying... [id=projects/cloud-486909/zones/europe-west1-b/instances/tp-nginx-vm, 10s elapsed]
google_compute_firewall.allow_web: Still destroying... [id=projects/cloud-486909/global/firewalls/allow-http, 10s elapsed]
google_compute_firewall.allow_ssh: Still destroying... [id=projects/cloud-486909/global/firewalls/allow-ssh, 10s elapsed]
google_compute_firewall.allow_web: Still destroying... [id=projects/cloud-486909/global/firewalls/allow-http, 20s elapsed]
google_compute_instance.vm_instance: Still destroying... [id=projects/cloud-486909/zones/europe-west1-b/instances/tp-nginx-vm, 20s elapsed]
google_compute_firewall.allow_ssh: Still destroying... [id=projects/cloud-486909/global/firewalls/allow-ssh, 20s elapsed]
google_compute_firewall.allow_ssh: Destruction complete after 22s
google_compute_firewall.allow_web: Destruction complete after 22s
google_compute_instance.vm_instance: Still destroying... [id=projects/cloud-486909/zones/europe-west1-b/instances/tp-nginx-vm, 30s elapsed]
google_compute_instance.vm_instance: Still destroying... [id=projects/cloud-486909/zones/europe-west1-b/instances/tp-nginx-vm, 40s elapsed]
google_compute_instance.vm_instance: Destruction complete after 41s
google_compute_subnetwork.subnet: Destroying... [id=projects/cloud-486909/regions/europe-west1/subnetworks/tp-nginx-subnet]
google_compute_address.static_ip: Destroying... [id=projects/cloud-486909/regions/europe-west1/addresses/tp-nginx-ip]
google_compute_address.static_ip: Still destroying... [id=projects/cloud-486909/regions/europe-west1/addresses/tp-nginx-ip, 10s elapsed]
google_compute_subnetwork.subnet: Still destroying... [id=projects/cloud-486909/regions/europe-west1/subnetworks/tp-nginx-subnet, 10s elapsed]
google_compute_address.static_ip: Destruction complete after 10s
google_compute_subnetwork.subnet: Still destroying... [id=projects/cloud-486909/regions/europe-west1/subnetworks/tp-nginx-subnet, 20s elapsed]
google_compute_subnetwork.subnet: Still destroying... [id=projects/cloud-486909/regions/europe-west1/subnetworks/tp-nginx-subnet, 30s elapsed]
google_compute_subnetwork.subnet: Destruction complete after 31s
google_compute_network.vpc: Destroying... [id=projects/cloud-486909/global/networks/tp-nginx-vpc]
google_compute_network.vpc: Still destroying... [id=projects/cloud-486909/global/networks/tp-nginx-vpc, 10s elapsed]
google_compute_network.vpc: Still destroying... [id=projects/cloud-486909/global/networks/tp-nginx-vpc, 20s elapsed]
google_compute_network.vpc: Destruction complete after 21s

Destroy complete! Resources: 6 destroyed.
yelizavetasamuilava@cloudshell:~/tp-nginx-gcp (cloud-486909)$
```

Plan

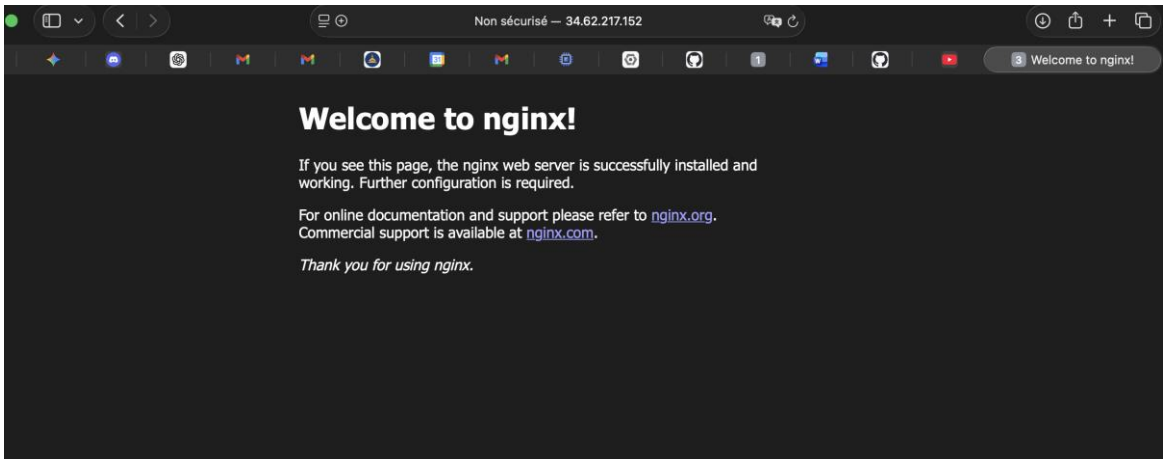
```
yelizavetasamuilava@cloudshell:~/tp-nginx-gcp (cloud-486909)$ terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # google_compute_address.static_ip will be created
  + resource "google_compute_address" "static_ip" {
      + address            = (known after apply)
      + address_type       = "EXTERNAL"
      + creation_timestamp = (known after apply)
      + effective_labels   = (known after apply)
      + id                 = (known after apply)
      + label_fingerprint  = (known after apply)
      + name               = "tp-nginx-ip"
      + network_tier       = (known after apply)
      + prefix_length      = (known after apply)
      + project            = "cloud-486909"
      + purpose            = (known after apply)
      + region             = (known after apply)
      + self_link          = (known after apply)
      + subnetwork         = (known after apply)
      + terraform_labels   = (known after apply)
      + users              = (known after apply)
    }

  # google_compute_firewall.allow_ssh will be created
  + resource "google_compute_firewall" "allow_ssh" {
      + creation_timestamp = (known after apply)
      + destination_ranges = (known after apply)
      + direction          = (known after apply)
      + enable_logging     = (known after apply)
      + id                 = (known after apply)
      + name               = "allow-ssh"
```

```
  # google_compute_subnetwork.subnet will be created
  + resource "google_compute_subnetwork" "subnet" {
      + creation_timestamp         = (known after apply)
      + external_ipv6_prefix       = (known after apply)
      + fingerprint                = (known after apply)
      + gateway_address            = (known after apply)
      + id                         = (known after apply)
      + internal_ipv6_prefix       = (known after apply)
      + ip_cidr_range              = "10.0.1.0/24"
      + ipv6_cidr_range            = (known after apply)
      + name                       = "tp-nginx-subnet"
      + network                    = (known after apply)
      + private_ip_google_access   = (known after apply)
      + private_ipv6_google_access = (known after apply)
      + project                    = "cloud-486909"
      + purpose                    = (known after apply)
      + region                     = "europe-west1"
      + secondary_ip_range         = (known after apply)
      + self_link                  = (known after apply)
      + stack_type                 = (known after apply)
    }

Plan: 6 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + public_ip = (known after apply)


Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
yelizavetasamuilava@cloudshell:~/tp-nginx-gcp (cloud-486909)$
```

terraform apply -auto-approve

```
yelizavetasamuilava@cloudshell:~/tp-nginx-gcp (cloud-486909)$ terraform apply -auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # google_compute_address.static_ip will be created
  + resource "google_compute_address" "static_ip" {
      + address            = (known after apply)
      + address_type       = "EXTERNAL"
      + creation_timestamp = (known after apply)
      + effective_labels   = (known after apply)
      + id                 = (known after apply)
      + label_fingerprint  = (known after apply)
      + name               = "tp-nginx-ip"
      + network_tier       = (known after apply)
      + prefix_length      = (known after apply)
      + project            = "cloud-486909"
      + purpose            = (known after apply)
      + region             = (known after apply)
      + self_link          = (known after apply)
      + subnetwork         = (known after apply)
      + terraform_labels   = (known after apply)
      + users              = (known after apply)
    }
```

```
google_compute_network.vpc: Creating...
google_compute_address.static_ip: Creating...
google_compute_network.vpc: Still creating... [10s elapsed]
google_compute_address.static_ip: Still creating... [10s elapsed]
google_compute_address.static_ip: Creation complete after 10s [id=projects/cloud-486909/regions/europe-west1/addresses/tp-nginx-ip]
google_compute_network.vpc: Creation complete after 11s [id=projects/cloud-486909/global/networks/tp-nginx-vpc]
google_compute_subnetwork.subnet: Creating...
google_compute_firewall.allow_web: Creating...
google_compute_firewall.allow_ssh: Creating...
google_compute_firewall.allow_ssh: Still creating... [10s elapsed]
google_compute_firewall.allow_web: Still creating... [10s elapsed]
google_compute_subnetwork.subnet: Still creating... [10s elapsed]
google_compute_firewall.allow_web: Creation complete after 11s [id=projects/cloud-486909/global/firewalls/allow-http]
google_compute_firewall.allow_ssh: Creation complete after 11s [id=projects/cloud-486909/global/firewalls/allow-ssh]
google_compute_subnetwork.subnet: Still creating... [20s elapsed]
google_compute_subnetwork.subnet: Creation complete after 21s [id=projects/cloud-486909/regions/europe-west1/subnetworks/tp-nginx-subnet]
google_compute_instance.vm_instance: Creating...
google_compute_instance.vm_instance: Still creating... [10s elapsed]
google_compute_instance.vm_instance: Still creating... [20s elapsed]
google_compute_instance.vm_instance: Still creating... [30s elapsed]
google_compute_instance.vm_instance: Creation complete after 31s [id=projects/cloud-486909/zones/europe-west1-b/instances/tp-nginx-vm]

Apply complete! Resources: 6 added, 0 changed, 0 destroyed.

Outputs:

public_ip = "34.62.217.152"
```

Ansible

```
yelizavetasamuilava@cloudshell:~/tp-nginx-gcp (cloud-486909)$ ansible-playbook -i "34.62.217.152," -u ansible --private-key ~/.ssh/id_ed25519 playbook.yml

PLAY [Installation de NGINX pour le TP] ***********************************************************************************************************

TASK [Gathering Facts] ****************************************************************************************************************************
[WARNING]: Host '34.62.217.152' is using the discovered Python interpreter at '/usr/bin/python3.11', but future installation of another Python interpreter could cau
se a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [34.62.217.152]

TASK [Mise a jour du cache APT] ********************************************************************************************************************
changed: [34.62.217.152]

TASK [Installer NGINX] ****************************************************************************************************************************
changed: [34.62.217.152]

TASK [Demarrer et activer NGINX] ******************************************************************************************************************
ok: [34.62.217.152]

PLAY RECAP ****************************************************************************************************************************************
34.62.217.152              : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```